

Experimental Methods Supplement

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The probability and decision rules use the measurable-event conditions of main Section 2.0 and Mathematical Supplement S19. In particular, every recorded capacity diagnostic, cell tag, coverage event and PASS decision is measurable under the declared data law.

Article: *Causal Foundations and Experimental Certification of Contingent Assignment, Heredity, and Reflexive Maintenance*

This supplement specifies how a physical experiment could assess the mathematical conditions. It gives a prospective correspondence and validation method and reports no new laboratory results. Its sections cover selected evidence (E1), architecture requirements (E2), development and validation controls (E3), a worked split-ribozyme design (E4), and unresolved experimental work (E5). References use the same numbers as the main article.

The design labels S1-X and TX-A2/HHR-PC distinguish the specified architecture and its worked candidate; they do not denote experimentally validated systems. The C0-E/C0-P/C0-V/C1-F and V1-V7 labels refer to protocol stages and controls defined below. These stages distinguish exploratory development, independent validation, and confirmatory testing.

E1. Existing evidence: what current experiments do and do not establish

The evidence comparison asks which component mechanisms published systems demonstrate, which pre-declared diagnostic criteria they meet, and which hypotheses remain unestablished. It does not identify a historical origin mechanism.

In the selected, non-systematic literature set examined through August 2026, we did not identify a system shown to satisfy all the conditions studied here. This observation applies only to that selected set. It is not a systematic-review result, a nonexistence theorem, or a priority claim. A systematic review would require a prespecified search scope, databases, query families, inclusion and exclusion criteria, and a screening record.

The examples below distinguish evidence that a component can work from evidence that a system meets the assumptions of a theorem.

E1.1 Catalytic templating

Cabello-Garcia et al. demonstrated enzyme-free, sequence-specific catalytic DNA templating with weak product inhibition, high selectivity among several dimer products, turnover, downstream-network participation, and covalent bond formation. This strongly supports the physical feasibility of catalytic information propagation. For decoder-free-origin testing, however, the relevant recognition domains are rationally designed; the mapping is therefore present in the engineered architecture before the run. The experiment is a strong component precedent but does not satisfy the origin-neutrality requirement for the full conjunction studied here.

E1.2 Open-ended RNA replication

Attwater et al. demonstrated exponential replication of defined and mixed RNA sequences under coupled pH/freeze-thaw cycling and showed open-ended sequence-pool behavior. This establishes a strong replication and cycling scaffold. Replication alone, however, is not a contingent operational assignment and therefore cannot substitute for the assignment or assignment-specific heredity gates.

These experiments use a supplied polymerase ribozyme. The study includes both defined-primer experiments and primer-free amplification, but does not establish autonomous formation and complete self-replication of the polymerase ribozyme itself [18].

E1.3 Autocatalytic RNA networks and maintenance chemistry

Ameta et al. demonstrated multispecies autocatalytic RNA reaction networks inside coacervates. Radakovic et al. demonstrated autocatalytic assembly of a chimeric aminoacyl-RNA synthetase ribozyme that can persist across repeated serial dilution and assemble another ribozyme in trans. These systems

provide strong precedents for network support, autocatalysis, and persistence. They do not independently establish a prior decoder-free assignment relation followed by the specific future-capacity intervention required by DFI-R(cap).

E1.4 RNA–peptide and aminoacylation chemistry

Ghosh et al. showed that peptide sequence can strongly alter hammerhead-ribozyme kinetics inside RNA–peptide coacervates. Singh et al. demonstrated broad thioester-mediated RNA aminoacylation and peptidyl-RNA synthesis in water. These results materially strengthen the physical plausibility of RNA/peptide support chemistry and cross-domain coupling, but they do not by themselves establish contingent assignment genesis.

The Ghosh study also observed inhibition relative to buffer, so sequence-dependent modulation should not be read as uniformly enhanced catalysis or as an established positive maintenance effect for the proposed system [21].

E1.5 Compartment cycling

Tsui demonstrated repetitive liposome fusion coupled to RNA replication across at least seven cycles. This is a useful precedent for repeated compartment-level redistribution and replication. It is not an assignment-specific heredity assay.

The reported DMSO-assisted low-temperature content-exchange conditions include non-freezing operation and should not be conflated with the pH/freeze–thaw replication protocol above [27].

The general pattern is therefore:

many strong physical modules $\nRightarrow$ full theorem-class realization.

E2. Physical requirements for candidate implementations

E2.1 Initial feasibility criteria

Initial assessment tests whether the proposed architecture already contains the assignment relation whose formation is to be explained. Robustness and persistence cannot establish spontaneous assignment formation when that relation is supplied in the initial architecture.

The initial assessment addresses the following questions:

1. Is the origin relation genuinely neutral?
2. Can a contingent organization-dependent mapping arise without target-dependent retention?
3. Can organization be intervened on cleanly?
4. Can assignment-specific heredity be distinguished from reconstruction?
5. Can assignment-mediated output support future assignment capacity?
6. Can all certificates coexist in one compatible system?
7. Can finite implementation and resource conditions be bounded?

Existing platforms were separated into the published system “as is” and a prospective redesign that may borrow only physically demonstrated capabilities.

Membership checks for origin, heredity and support

An origin audit must justify DC0 from main Section 2.4, including detection of dormant stored relations within the declared ontology. A deletion that leaves all measured channels null does not by itself show that no relation was present. If a delayed or gate-opening diagnostic is used, establish that it exposes an origin organization while preventing new relation formation during the diagnostic word. Specify the state-indexed signature or an independently justified structural absence result; ensemble cancellation of randomly stored orientations is insufficient. Inadequate diagnostic access yields INCONCLUSIVE, while an established preinstalled relation fails origin membership.

For DFI-H, document all four surgical arms, almost-sure aligned loading, complete joint-context matching, and the common child kernel, including regime-wise child-noise/context separation in S4.0. Equal noise and context marginals alone do not establish that kernel.

For the fixed-state DFI-R(cap) target, assess RASM+ independently of the whole-output contrast. Matched natural organization regimes must load the same distinct support states used in the sham/block

comparison, under the specified common context and maintenance mechanism. A carrier with an assignment-dependent reporting coordinate and an unrelated constant housekeeping coordinate cannot pass this condition merely by showing a positive support-block effect. This article reports no laboratory test establishing that requirement.

For UFE, record the matched-state event and use its conditional context law in both primitive defects. The transcript must determine the tested joint response and conditioning cell, whose realized support is positive. Missing support or a deficient transcript makes that conditional assessment unassessable.

E2.2 Dominant bottleneck

Among the architectures considered, the dominant bottleneck was not generic catalysis, replication, or compartmentalization. It was the transition

auditable assignment-neutral origin $\rightarrow$ genuine contingent organization-dependent mapping.

Designed templating platforms fail early because their recognition relation is already encoded. Replication platforms often reach downstream persistence without ever defining the required assignment object.

E2.3 A conditional exchangeable-node routing design

S1-X is a proposed synthetic architecture for a proof-of-principle test of this transition. Its components and protocol requirements are:

- two chemically identical input-token copies;
- two chemically identical output-token copies;
- a pre-genesis generative protocol intended to be invariant under within-domain permutation;
- one generic monovalent cross-domain coupling chemistry;
- pair-blind stochastic matching as a physical condition to be validated, not inferred from linker identity alone;
- one generic link-local transduction mechanism;
- post-genesis token tracking and challenge through an admissible non-mapping addressability layer;
- no pair-specific sequence tuning.

Within-domain chemical identity of the copies is a membership requirement of the specified S1-X architecture. It is not asserted to be necessary for exchangeability in general, and it is not sufficient for exchangeability of the complete generative protocol. The relevant symmetry is a property of the complete pre-genesis generative protocol, including preparation, spatial placement, labeling, illumination history, timing, handling, coupling opportunities, and randomization. Empirical approximate equality cannot create exact exchangeability after the fact; origin-symmetry checks, V6/V7, and later mapping-orientation controls can only detect implementation violations of the declared generative symmetry.

Token indices are observational bookkeeping only and do not enter the dynamics. Individual addressability must be obtained either through exchangeable spatial/world-line observation or through a post-genesis tracking layer that neither selects the pairing nor writes the output and that passes perturbation/symmetry controls. If individual-token tracking requires copy-specific pre-genesis chemistry that changes the dynamics, strict S1-X membership is not established.

Accordingly, S1-X is a **conditional implementation candidate**: it avoids a preinstalled pair-specific mapping provided that the full preparation/joining protocol is permutation-invariant within domains and that an admissible non-mapping tracking/addressability layer exists. It does not establish that a particular molecular construct works.

E3. Separation of development, validation, and confirmation

E3.1 Global calibration and pair-specific optimization

Real molecular engineering requires optimization. Forbidding all calibration would make the framework experimentally unusable. Allowing unrestricted calibration would let the experimenter reconstruct the desired mapping by design.

The solution is to separate **global component engineering** from **pair-specific mapping optimization**.

A global parameter vector may include:

$$\theta_{\text{global}} = (\theta_{\text{transducer}}, \theta_{\text{arm}}, \theta_{\text{link}}, \theta_{\text{challenge}}, \theta_{\text{reporter}}, \theta_{\text{ablation}}, \theta_{\text{buffer}}, \theta_{\text{time}}).$$

These parameters may be optimized in non-contingency systems, provided the same prespecified values are later used for every input/output copy and every possible pairing.

Pair-dependent tuning is forbidden.

E3.2 Four stages

The calibration sequence is:

$$C0-E \rightarrow C0-P \rightarrow C0-V \rightarrow C1-F.$$

- **C0-E:** exploratory global engineering.
- **C0-P:** provisional construct and validation preregistration, including the experimental unit, sample size per arm, batch/lot allocation, randomization/blinding where applicable, stopping rule, missingness/exclusion rules, batch-rejection rule, analysis rule, and any predeclared sequential-design procedure.
- **C0-V:** independent non-contingency validation.
- **C1-F:** final specification of the validated implementation before confirmatory testing.

A functional random 2×2 contingent-mapping experiment is forbidden before C1-F.

After C0-P, outcome-dependent extension of C0-V that was not covered by the prespecified analysis invalidates the stated validation guarantee. Further modification or sampling must be treated as development and followed by new independent validation. The default fixed-sample design declares the number of experimental units per arm and analyzes once at the declared end, with no outcome-dependent early PASS or extra sampling. A sequential alternative must instead name a confidence-sequence or stopping-specific calibration whose familywise error guarantee has been established for its actual cell streams and stopping rule. Its error allocation, stopping boundaries, and maximum information horizon are fixed before validation begins. Preregistration alone does not establish validity. Section E3.5 specifies the corresponding acquisition and power requirements.

E3.3 Why 1×1 calibration is insufficient

A 1×1 linked system can show that a transducer works, but it cannot detect broadcast to a second output or challenge spillover to another input.

The validation matrix therefore contains:

- **V1:** unlinked 1×1 , pre-link background;
- **V2:** linked 1×1 , post-link function;
- **V3:** linked 1×2 , linked-versus-off-output routing;
- **V4:** linked 2×1 , linked-versus-off-input addressability;
- **V5:** linked 1×1 , organization ablation versus sham;
- **V6:** tracking perturbation;
- **V7:** monovalency/aggregate control;
- **H0:** inert apparatus-level symmetry/handling control.

These controls deliberately fix the link relation. They assess calibration and measurement performance, not spontaneous formation of a contingent assignment.

V1–V7 are necessary component-validation tests, not a proof that the complete functional 2×2 context contains no higher-order interaction. The prospective confirmatory 2×2 protocol must therefore include preregistered joint-context artifact/bypass controls for effects such as resource competition, higher-order aggregation, reporter saturation, or full-context interactions. Detection of an unmodelled full-context interaction capable of generating the routing contrast prevents PASS under the prespecified structural exclusion criteria.

E3.4 Prespecified validation decision

For normalized response $R \in [0, 1]$, one conservative confidence radius for statistically valid independent bounded experimental units is

$$h(n, \alpha) = \sqrt{\frac{\log(2/\alpha)}{2n}}.$$

Here n denotes the number of valid experimental units under the preregistered dependence model, not

the number of raw frames, technical repeats, time points, wells split from one preparation, or nested observations. Clustered, repeated-measures, lineage-dependent, batch-dependent, or adaptive data require an inferential/concentration procedure valid for that dependence structure. The Hoeffding expression is therefore an illustrative conservative device, not a universal or optimal statistical prescription.

For a two-arm contrast

$$\widehat{\Delta} = \bar{R}_1 - \bar{R}_0,$$

a conservative confidence radius can be formed from the sum of the two arm-wise radii after allocation of the calibration confidence budget.

Let α_{cal} denote the familywise calibration confidence budget. Before C0-V, allocate $\alpha_k > 0$ to the required quantitative inequalities so that

$$\sum_k \alpha_k \leq \alpha_{\text{cal}}.$$

Individual intervals use their preallocated budgets, while final calibration certification is conjunctive: every required hard gate and quantitative inequality must pass. This differs from selecting one significant result from many tests and does not authorize post-hoc reallocation of α_k .

Before C0-V begins, the protocol specifies quantities such as:

$$\varepsilon_{\text{pre}}^*, \quad m_{\text{route}}^*, \quad m_{\text{addr}}^*, \quad m_A^*,$$

as well as tracking, aggregation, sham, and illumination tolerances.

A construct advances to C1-F only if every required hard gate and quantitative inequality passes simultaneously. A failed validation can return to development only under a new construct version and new preregistration. Data used to revise the construct are development data and cannot also serve as independent validation of the revised construct.

The stated α_{cal} guarantee is for one specified construct and its prespecified validation design. Returning to development and testing new versions does not give the entire development history the same error bound. A claim controlling erroneous certification across multiple versions requires a separately declared cross-version error budget or a valid final-validation and retry policy. No such whole-development probability claim is made here.

E3.5 Sample adequacy, acquisition, and power

The statistical error bound and the chance of obtaining PASS answer different questions. The following requirements separate confidence validity, acquisition of usable data, and power. IID abbreviates independent and identically distributed.

If the registered target is the capacity-strengthened conjunction of main Section 7.3, its target vector must contain both the support-loop and actual-capacity diagnostics, with separate retained floors and adequate erosion/precision budgets as in main Section 9.6. A capacity-only replacement target must be labelled separately; it does not establish the conjunction.

Before validation, record whether the study promises only error-controlled certification or also a lower bound on its probability of obtaining PASS under a declared alternative. The former needs valid simultaneous confidence bounds; the latter also needs justified separation margins and sufficient precision. Sample-size preregistration alone establishes neither power nor physical-system membership.

For a bounded two-arm response contrast, let $r_j = h(n_j, \alpha_j)$ at the valid fixed sample sizes. The lower confidence bound is $\widehat{\Delta} - r_0 - r_1$. If the true contrast exceeds its required threshold by at least $\gamma > 0$, the sufficient planning inequality $2(r_0 + r_1) < \gamma$ ensures that simultaneous coverage certifies that contrast. Apply the corresponding upper/lower-bound calculation to every quantitative condition, using the full familywise calibration budget. If such separation assumptions are unverified, do not report a guaranteed PASS probability.

Predeclare the experimental unit, cell/tag definition, minimum usable count for each required condition, missingness and exclusion mechanisms, and the event of adequate acquisition. Under fixed-size IID tagged sampling, the Mathematical Supplement, S13.3, justifies using realized within-cell counts by conditioning on the complete tag vector. Other dependence or selection structures require an appropriate procedure. A zero or inadequate required count yields INCONCLUSIVE for that quantitative requirement; an independently established hard-condition failure retains its failure status. No response-dependent deletion or

unplanned sampling extension can turn this into PASS.

The acquisition-failure bound concerns obtaining the declared usable counts; it must follow from the design and a justified cell-probability model. Coverage plus acquisition controls their joint event. A PASS-power guarantee further requires that this event establishes every hard and quantitative requirement, as in the Mathematical Supplement, S13.2. When sequential validation is chosen, the confidence bounds must cover all allowed looks or have an independently proved stopping-specific calibration; the scalar Hoeffding radius above with one unchanged budget is not reused at arbitrary stopping times. The same requirements apply to the later confirmatory test with its separately allocated error budget.

No actual sample size or achieved power is claimed for TX-A2/HHR-PC: the required calibration measurements and separation margins are not yet available.

E4. Worked pre-binding architecture: TX-A2/HHR-PC

This worked case illustrates a prospective design. It reports no experimental result.

The selected pre-binding family is:

TX-A2/HHR-PC: photo-gated, link-completed split-hammerhead RNA router.

The conceptual input token contains one common photo-gatable hammerhead/transducer half and one universal input-side coupling handle. The output token contains one common complementary hammerhead/transducer half, an output-local fluorogenic cleavage readout, and one universal output-side coupling handle. The two copies within each domain are chemically identical.

The generic link is intended to perform two roles: it selects which output is downstream of a given input, and it co-localizes transducer halves so that the same local transduction logic can operate on every realized pair. This interpretation remains conditional on the complete joining protocol being pair-blind in its physical opportunity distribution. The challenge is delivered after the genesis cut to one tracked input token using the same intervention chemistry available to either input. The output is read locally at the linked versus off-output token, conditional on an admissible tracking/addressability layer that is non-mapping, nonselective, and perturbatively controlled.

Distinct component precedents support different parts of the prospective design. Gambill et al. [20] demonstrate split Tetrahymena group-I ribozyme complementation coupled to protein outputs, not a split-hammerhead router. Kertsburg and Soukup [22] provide an allosteric communication-module precedent, while Martick et al. [23] establish activity of a discontinuous hammerhead within a messenger RNA. Lim et al. [24] provide a programmable RNA-triggered signal-transduction precedent based on a hairpin ribozyme. Optical control of hammerhead activity has also been demonstrated using photoreversible RNA acylation [35]. These results were obtained in different constructs and do not validate the proposed link-local split-hammerhead architecture. The generic joining chemistry, output-local fluorogenic readout, non-mapping addressability and compatibility of all components remain implementation requirements to be established for this design.

The proposed architecture requires a permutation-invariant preparation and joining protocol, together with a tracking and addressing method that neither selects pairings nor writes the output. Existing studies support individual component mechanisms, but an integrated working construct has not been established. No implementation has been validated for the proposed confirmatory random 2×2 experiment, and no experimental realization of the complete hypothesis class is reported.

E5. Residual Experimental Agenda and Open Implementation Questions

The implementation problems below remain experimentally unresolved. Neither the mathematical results nor the correspondence method assumes that they have been solved.

E5.1 Immediate unresolved calibration problem

The proposed implementation remains unfixed pending the required measurements. A future laboratory program would need to determine, using global rather than pair-specific engineering:

- an exact working input-token construct;

- an exact working output-token construct;
- split-hammerhead/transducer geometry;
- generic cross-domain coupling placement and linker geometry;
- photoactivation chemistry and spatial-resolution method;
- local fluorogenic output readout;
- generic organization-ablation chemistry;
- buffer, ionic conditions, temperature, and observation windows.

The unresolved measured quantities include:

- pre-link challenge-dependent background;
- post-link functional gain;
- linked-versus-off-output crosstalk;
- linked-versus-off-input addressability;
- organization-ablation specificity;
- tracking perturbation;
- aggregate/multivalent fraction;
- between-run and between-batch variability.

E5.2 Required development sequence

A strict realization should proceed:

$$C0-E \rightarrow C0-P \rightarrow C0-V \rightarrow C1-F \rightarrow \text{prospective functional } 2 \times 2 \text{ test.}$$

Here C0-E denotes exploratory global engineering; C0-P specifies and preregisters the validation protocol; C0-V is independent non-contingency validation; and C1-F fixes the validated implementation before confirmatory testing.

A functional random 2×2 mapping outcome should not be used during development.

E5.3 Prospective full test

Only after C1-F would a random 2×2 theory-conformant test become eligible for preregistration. Such a test would need to retain all realized mapping orientations, challenge arbitrary tracked input tokens after genesis, quantify linked versus off-output responses, preserve the predeclared origin, intervention, exclusion, sample-size, and stopping rules, and include preregistered joint-context artifact/bypass controls capable of detecting full- 2×2 -only interactions that could mimic routing.

Passing these initial origin and assignment tests would still not complete the whole theory. Later prospective work would need assignment-specific heredity, actual-future-capacity reflexive maintenance, single-system composition, finite-error certification, robustness, and explicit resource accounting.

E5.4 Stronger molecular and prebiotic question

The exchangeable-node S1-X architecture is a synthetic theory-conformance candidate. A stronger question remains open:

Can a molecularly natural or prebiotically plausible system realize the same stated hypothesis class without engineered symmetric bookkeeping or a deliberately constructed routing scaffold?

This stronger question should not be answered by retroactively weakening the theorem or by treating approximate chemical similarity as exact symmetry. It may require a separate correspondence class, new experiments, or a future mathematical extension explicitly declared outside the mathematical scope of this article.

E5.5 Invitation to falsify

The most useful future outcome need not be a successful realization. A decisive demonstration that every physically credible implementation of the declared S1-X family requires pair-specific tuning, unavoidable broadcast, origin asymmetry, or destructive causal intervention would exclude that implementation family under the stated criteria.

The experimental aim is to determine whether a physical implementation satisfies the stated assumptions. A reproducible violation of a required structural condition would exclude that implementation from the hypothesis class.

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